CH1101: Quantum Chemistry and Spectroscopy

Tutorial Sheet 07

Module: Quantum Chemistry

Q1. Using the linear combinations of atomic orbitals (LCAO), obtain wavefunctions for the bonding and anibonding molecular orbitals (MOs) of H_2^+ molecule.

Q2. A homonuclear diatomic system has the ground-state molecular orbital (MO) configuration $1\sigma_g^2 1\sigma_u^2 2\sigma_g^2 2\sigma_u^2 3\sigma_g^2 1\pi_u^4 1\pi_g^2$. Answer the following:

(a) What is the number of bonding electrons?

(b) Find the bond order.

(c) What would you expect the effect to be on the bond dissociation energy of this molecule when an electron is removed (a) from the $1\pi_g$ MO ? (b) from the $3\sigma_g$ MO?

Q3. Indicate whether you expect each of the following homonuclear diatomic MOs to be bonding or antibonding. Sketch the MO in each case:

(a) σ_u (b) π_u (c) π_g

Q4. Arrange bond dissociation energies of O_2, N_2, O_2^+, and N_2^+ in increasing bond order.

Q5. A homonuclear diatomic MO of π_u symmetry is to be expressed as a linear combination of AOs centered on the nuclei laying on the z-axis. Which of the AOs in the following list can contribute to the MO?

(a) 1s (b) 2s (c) $2p_x$ (d) $2p_y$ (e) $2p_z$ (f) $3p_x$ (g) $3p_y$ (h) $3p_z$